Audio Engine

User Manual

GD32 DSP Audio Playback System
for microcontrollers with I2S support
Version 2.0

`8-bit | 16-bit | Mono | Stereo | Runtime DSP | No FPU`

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Audio Engine User Manual

Overview

The Audio Engine is a reusable, embedded DSP audio playback system designed for GD32 microcontrollers with I2S support with audio output to digital amplifiers such as the MAX98357A.

Key Features

- **Dual Format Support**: 8-bit and 16-bit audio playback
- **Flexible Modes**: Mono and stereo playback
- **DSP Filter Chain**: Runtime-configurable filters with fixed-point arithmetic
- **No FPU Required**: All DSP operations use integer math for MCU efficiency
- **Sample Rate**: Default 22 kHz (configurable)
- **DMA-Driven**: Efficient I2S streaming with double-buffering
- **Low Latency**: ~93 ms playback latency with 2048-sample buffer

Core Capabilities

Feature	Specification
Sample Rates	22 kHz (default), configurable
Audio Depths	8-bit unsigned, 16-bit signed
Channels	Mono, Stereo
Buffer Size	2048 samples (ping-pong DMA)
Nyquist Frequency	11 kHz @ 22 kHz sample rate
Volume Control	Software configurable (0-3x gain)
Fade Effects	In/Out (~93 ms at 22 kHz)

Quick Start

1. Initialize the Audio Engine

You **must** call `AudioEngine_Init()` to set up the audio engine with the required hardware callbacks. This function initializes all filter state and validates that the necessary hardware interface functions are provided:

```
#include "audio_engine.h"


// Step 1: I2S peripheral clock/pins must be initialized in application startup
// (Project uses spi_config(speed) as the I2S re-init callback)

```

```
// Step 2: Initialize the audio engine with hardware callbacks
PB_StatusTypeDef status = AudioEngine_Init(
 DAC_MasterSwitch, // Function to control amplifier on/off
 ReadVolume, // Function to read volume setting
 spi_config // Function to re-initialize I2S for a requested sample rat
 e
);

if( status != PB_Idle ) {
 // Handle initialization error
}

// Step 3: Configure filters (optional, defaults are pre-set)
SetLpf16BitLevel(LPF_Soft); // Set filter aggressiveness
SetFilterConfig(&my_filter_config); // Apply complete config

// Audio engine is now ready to play samples
```

2. Play a 16-bit Audio Sample

```
// Assuming 'doorbell_sound' is a 16-bit mono WAV sample in flash memory
// 44,100 samples = ~2 seconds @ 22 kHz, 16-bit mono

extern const uint8_t doorbell_sound[];
extern const uint32_t doorbell_sound_size;

PB_StatusTypeDef result = PlaySample(
 doorbell_sound, // Pointer to audio data
 doorbell_sound_size, // Total samples (all channels combined)
 22000, // Sample rate (Hz)
 16, // Bit depth (16 = 16-bit)
 Mode_stereo // Stereo playback
);

// Wait for playback to complete
WaitForSampleEnd();
```

3. Configure Filters

```
// Get current filter configuration
FilterConfig_TypeDef cfg;
GetFilterConfig(&cfg);

// Adjust filter levels
cfg.enable_16bit_biquad_lpf = 1; // Enable 16-bit LPF
cfg.enable_soft_clipping = 1; // Enable soft clipping to prevent distortion
cfg.lpf_16bit_level = LPF_Medium; // Medium filtering strength

// Apply new configuration
SetFilterConfig(&cfg);

// Or use convenience function for LPF level
SetLpf16BitLevel(LPF_Aggressive); // Stronger filtering

// 8-bit path: set a custom LPF alpha and read it back
SetLpf8BitLevel(LPF_Custom);
SetLpf8BitCustomAlpha(CalcLpf8BitAlphaFromCutoff(900.0f, 11025.0f));
uint16_t current_alpha = GetLpf8BitCustomAlpha();
```

Architecture

System Block Diagram

The audio playback system follows a clear data flow from flash memory through DSP processing to the I2S output. The DMA operates in ping-pong mode, processing audio chunks as they're transmitted, ensuring continuous playback without CPU blocking.

Filter Chain Stages (16-bit Audio)

1. **Biquad Low-Pass Filter** *(Optional - enable_16bit_biquad_lpf)*
   - Second-order IIR filter
   - Runtime-configurable aggressiveness: Very Soft → Aggressive
   - Warm-up: 16 passes of first sample to prevent startup artefacts
   - Cutoff range (22 kHz fs, approx): ~2.6 kHz (Very Soft), ~1.4 kHz (Soft), ~0.9 kHz (Medium), ~0.2 kHz (Aggressive)
   - 64-bit accumulator in the biquad path to prevent overflow with aggressive settings

2. **DC Blocking Filter** *(Selectable - enable_soft_dc_filter_16bit)*
   - Removes DC offset and very low frequencies
   - Two variants: standard (44 Hz) or soft (22 Hz)
   - Prevents output drift
   - Always active in one of the two modes

3. **Air Effect (High-Shelf)** *(Optional - enable_air_effect)*
   - Adds presence and brightness to audio
   - Runtime-adjustable boost (+1 dB, +2 dB, +3 dB presets)
   - Disabled by default

4. **Fade In/Out** *(Enabled by default)*
   - Quadratic power curve ramp
   - Default: 2048 samples (~93 ms @ 22 kHz)
   - Smooth entry/exit for audio transitions
   - Can be disabled via `SetFadersEnabled(0)` if needed

5. **Noise Gate** *(Optional - enable_noise_gate)*
   - Mutes samples below ±512 amplitude
   - Suppresses quantization noise during silence
   - Disabled by default

6. **Soft Clipping** *(Optional - enable_soft_clipping)*
   - Smooth cubic curve limiting above ±28,000
   - Prevents harsh digital clipping

- Musical, natural-sounding compression
- Recommended: keep enabled

7. **Volume Scaling** *(Always Active)*
    - Integer multiplication (0–3x gain)
    - Read from hardware GPIO (3-level selector)
    - Applied per-sample

Filter Chain Stages (8-bit Audio)

1. **8-bit to 16-bit Conversion with Dithering** *(Always Active)*
    - TPDF (Triangular PDF) dithering reduces quantization noise
    - Upsamples to internal 16-bit working format

2. **One-Pole Low-Pass Filter** *(Optional - enable_8bit_lpf)*
    - One-pole IIR (alpha range: 0.625 to 0.9375)
    - Separate aggressiveness levels for 8-bit audio
    - Custom alpha supported via `SetLpf8BitCustomAlpha()` and `CalcLpf8BitAlphaFromCutoff()`
    - Current custom alpha can be read with `GetLpf8BitCustomAlpha()`

3. **Makeup Gain** *(Always Active when LPF enabled)*
    - Post-LPF amplitude compensation (~1.08x default)
    - Configurable via `SetLpfMakeupGain8Bit()`

4. **DC Blocking & Remaining Stages**
    - Same as 16-bit path (steps 2–7)
    - Air Effect, Fade, Noise Gate, Soft Clipping, Volume Scaling

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API Reference

Enumeration Types

`PB_StatusTypeDef`

Audio playback state enumeration.

```
typedef enum {
 PB_Idle, // No audio playing
 PB_Error, // Error during playback
 PB_Playing, // Audio actively playing
 PB_Pausing, // Fade-out in progress (pause or stop)
```

```
 PB_Paused, // Playback paused
 PB_PlayingFailed // Playback failed to start
} PB_StatusTypeDef;
```

`PB_ModeTypeDef`

Playback channel mode.

```
typedef enum {
 Mode_stereo, // Stereo (2-channel) playback
 Mode_mono // Mono (single-channel) playback
} PB_ModeTypeDef;
```

`LPF_Level`

Low-pass filter aggressiveness level for 16-bit and 8-bit LPFs.

```
typedef enum {
 LPF_Off, // Filtering disabled
 LPF_VerySoft, // Minimal filtering (α = 0.625)
 LPF_Soft, // Gentle filtering (α ≈ 0.80)
 LPF_Medium, // Balanced filtering (α = 0.875)
 LPF_Firm, // Firm filtering (α ≈ 0.92)
 LPF_Aggressive, // Strong filtering (α ≈ 0.97)
 LPF_Custom // Use custom alpha (SetLpf16BitCustomAlpha or SetLpf8BitCustomA
 lpha)
} LPF_Level;
```

Structure Types

`FilterConfig_TypeDef`

Runtime filter configuration structure.

```
typedef struct {
 uint8_t enable_16bit_biquad_lpf; // Enable/disable 16-bit biquad LPF
 uint8_t enable_soft_dc_filter_16bit; // Use softer DC blocking (22 Hz vs 44 Hz)
 uint8_t enable_8bit_lpf; // Enable/disable 8-bit one-pole LPF
 uint8_t enable_noise_gate; // Enable/disable noise gate
 uint8_t enable_soft_clipping; // Enable/disable soft clipping
 uint8_t enable_air_effect; // Enable/disable air effect
 uint32_t lpf_makeup_gain_q16; // Post-LPF gain in Q16 fixed-point
 uint32_t lpf_makeup_gain_16bit_q16; // Post-16-bit LPF gain in Q16 fixed-point
 LPF_Level lpf_16bit_level; // 16-bit LPF aggressiveness
 uint16_t lpf_16bit_custom_alpha; // Custom alpha for 16-bit LPF
 LPF_Level lpf_8bit_level; // 8-bit LPF aggressiveness
 uint16_t lpf_8bit_custom_alpha; // Custom alpha for 8-bit LPF
 uint8_t enable_filter_chain_16bit; // Master enable for entire 16-bit filter chain
```

```
 uint8_t enable_filter_chain_8bit; // Master enable for entire 8-bit filter chain
} FilterConfig_TypeDef;
```

Field Descriptions:

- `lpf_makeup_gain_q16` : Gain value in Q16 format (65536 = 1.0x). Default: 70779 (~1.08x)
- `lpf_16bit_level` : Filter aggressiveness affects cutoff frequency and stop-band attenuation
- `lpf_16bit_custom_alpha` : Custom alpha used when `lpf_16bit_level = LPF_Custom`
- `lpf_8bit_level` : Filter aggressiveness for the 8-bit one-pole LPF
- `lpf_8bit_custom_alpha` : Custom alpha used when `lpf_8bit_level = LPF_Custom`

`AudioEngine_HandleTypeDef`

Audio engine state handle (for initialization).

```
typedef struct {
 void *i2s_context; // Optional application-specific I2S context pointer
 int16_t *pb_buffer; // Playback buffer (2048 samples)
 uint32_t playback_speed; // Default playback speed (Hz)
} AudioEngine_HandleTypeDef;
```

Function Reference

Hardware Setup (Done in main.c)

Before playing audio, ensure:

1. **I2S is configured** in application startup code (22 kHz, 16-bit, DMA enabled)
2. `AudioEngine_Init()` **is called** with function pointers for:
   - DAC on/off control
   - Volume reading
   - I2S re-initialization

3. **Filters are configured** to desired settings (optional; defaults are applied by `AudioEngine_Init()`)

Calling `AudioEngine_Init()` is **required** before any audio playback. It initializes the filter state, validates hardware callbacks, and sets up default filter configuration.

Playback Control

`AudioEngine_Init()`

Initialize the audio engine with required hardware callbacks.

```
PB_StatusTypeDef AudioEngine_Init(
 DAC_SwitchFunc dac_switch,
 ReadVolumeFunc read_volume,
 I2S_InitFunc i2s_init
);
```

Parameters:

- `dac_switch` : Function pointer for controlling amplifier on/off (GPIO control)
- `read_volume` : Function pointer for reading current volume level
- `i2s_init` : Function pointer for I2S peripheral re-initialization

Returns:

- `PB_Idle` if initialization successful
- `PB_Error` if any function pointer is NULL

Important Notes:

- **Must be called once** before any call to `PlaySample()` or other playback functions
- Initializes all filter state variables and resets playback status
- Sets up default filter configuration (can be overridden with `SetFilterConfig()`)
- Validates that all required hardware callbacks are provided
- Does not start audio playback itself

Example:

```
#include "audio_engine.h"

// Define these functions in your application
void DAC_MasterSwitch(uint8_t state) {
 if (state) {
 gpio_bit_write(GPIOC, GPIO_PIN_0, SET); // Enable amplifier
 } else {
 gpio_bit_write(GPIOC, GPIO_PIN_0, RESET); // Disable amplifier
 }
}

uint16_t ReadVolume(void) {
 // Return volume level 1-65535
 return volume_setting;
}

// In main.c initialization:
PB_StatusTypeDef status = AudioEngine_Init(
 DAC_MasterSwitch,
 ReadVolume,
 spi_config
);

if (status != PB_Idle) {
 printf("Audio engine initialization failed!\n");
```

```
 return;
}

// Now safe to call PlaySample()
```

`PlaySample()`

Start playback of an audio sample.

```
PB_StatusTypeDef PlaySample(
 const void *sample_to_play,
 uint32_t sample_set_sz,
 uint32_t playback_speed,
 uint8_t sample_depth,
 PB_ModeTypeDef mode
);
```

Parameters:

- `sample_to_play` : Pointer to audio data in flash/RAM
- `sample_set_sz` : Total samples (all channels combined)
- `playback_speed` : Sample rate in Hz (typically 22000)
- `sample_depth` : 8 or 16 (bits per sample)
- `mode` : `Mode_mono` or `Mode_stereo`

Returns:

- `PB_Playing` if playback started successfully
- `PB_Error` or `PB_PlayingFailed` on error

Important Notes:

- Audio data is accessed in real-time during playback (must be in accessible memory)
- DMA directly reads from the provided buffer
- For 16-bit mono audio: `sample_set_sz = num_samples`
- For 16-bit stereo (interleaved): `sample_set_sz = 2 * num_frames`
- For 8-bit mono audio: `sample_set_sz = num_samples`
- For 8-bit stereo (interleaved): `sample_set_sz = 2 * num_frames`
- Blocks briefly while starting DMA
- Configure filters separately with `SetLpf16BitLevel()` or `SetFilterConfig()`

Example:

```
extern const uint8_t alert_sound_16bit_mono[];
extern const uint32_t alert_sound_16bit_mono_size;

SetLpf16BitLevel(LPF_Medium); // Medium filtering
```

```
PB_StatusTypeDef result = PlaySample(
 alert_sound_16bit_mono,
 alert_sound_16bit_mono_size,
 22000, // Sample rate
 16, // 16-bit
 Mode_mono
);

if (result != PB_Playing) {
 // Handle error
 printf("Playback failed: %d\n", result);
}
```

`WaitForSampleEnd()`

Block until audio playback completes.

```
PB_StatusTypeDef WaitForSampleEnd(void);
```

Returns:

- `PB_Idle` when playback finishes
- `PB_Error` if playback was interrupted

Example:

```
SetLpf16BitLevel(LPF_Soft);
PlaySample(my_sound, my_sound_size, 22000, 16, Mode_mono);
WaitForSampleEnd(); // Wait until done
printf("Playback complete\n");
```

`PausePlayback()`

Pause ongoing playback (can resume later) with smooth fade-out.

```
PB_StatusTypeDef PausePlayback(void);
```

Returns:

- `PB_Paused` on success
- `PB_Idle` if no audio was playing

Notes:

- Pause fadeout duration set by `SetPauseFadeTime()`
- Intelligently handles edge cases to prevent audible artefacts:
- **Pausing during fade-in:** Scales pause fadeout proportionally to start from current volume
- **Pausing during end-of-file fadeout:** Scales pause fadeout to maintain smooth volume continuity
- All transitions use quadratic volume curves for smooth audio

Example:

```
if (user_pressed_pause_button) {
 PausePlayback();
}
```

`ResumePlayback()`

Resume previously paused audio with smooth fade-in.

```
PB_StatusTypeDef ResumePlayback(void);
```

Returns:

- `PB_Playing` on success
- `PB_Idle` if no paused audio

Notes:

- Resume fadein duration set by `SetResumeFadeTime()`
- Audio fades in smoothly from silence using a quadratic volume curve
- Playback resumes from the exact position where it was paused

Example:

```
if (user_pressed_play_button && prev_state == PB_Paused) {
 ResumePlayback();
}
```

Filter Configuration

`SetFilterConfig()`

Apply a complete filter configuration.

```
void SetFilterConfig(const FilterConfig_TypeDef *cfg);
```

Parameters:

- `cfg` : Pointer to filter configuration structure

Example:

```
FilterConfig_TypeDef cfg = {
 .enable_16bit_biquad_lpf = 1,
 .enable_soft_dc_filter_16bit = 1,
 .enable_8bit_lpf = 1,
 .enable_noise_gate = 0,
 .enable_soft_clipping = 1,
 .enable_air_effect = 0,
```

```
  .lpf_makeup_gain_q16 = 70779,
  .lpf_16bit_level = LPF_Medium,
  .lpf_16bit_custom_alpha = LPF_16BIT_MEDIUM,
  .lpf_8bit_level = LPF_Medium,
  .lpf_8bit_custom_alpha = LPF_MEDIUM
};
SetFilterConfig(&cfg);
```

`GetFilterConfig()`

Read current filter configuration.

```
void GetFilterConfig(FilterConfig_TypeDef *cfg);
```

Example:

```
FilterConfig_TypeDef cfg;
GetFilterConfig(&cfg);
printf("LPF Level: %d\n", cfg.lpf_16bit_level);
```

`SetLpf16BitLevel()`

Change 16-bit LPF aggressiveness (convenience function).

```
void SetLpf16BitLevel(LPF_Level level);
```

Parameters:

- `level` : LPF_VerySoft, LPF_Soft, LPF_Medium, LPF_Firm, or LPF_Aggressive

Example:

```
SetLpf16BitLevel(LPF_Aggressive); // Strong filtering
```

`SetLpf8BitCustomAlpha()`

Set custom alpha for the 8-bit one-pole LPF.

```
void SetLpf8BitCustomAlpha(uint16_t alpha);
```

Example:

```
SetLpf8BitLevel(LPF_Custom);
SetLpf8BitCustomAlpha(LPF_MEDIUM);
```

`GetLpf8BitCustomAlpha()`

Read the current 8-bit LPF custom alpha value.

```
uint16_t GetLpf8BitCustomAlpha(void);
```

`CalcLpf8BitAlphaFromCutoff()`

Compute 8-bit LPF alpha from cutoff and sample rate.

```
uint16_t CalcLpf8BitAlphaFromCutoff(float cutoff_hz, float sample_rate_hz);
```

Example:

```
uint16_t alpha = CalcLpf8BitAlphaFromCutoff(4500.0f, 11000.0f);
SetLpf8BitLevel(LPF_Custom);
SetLpf8BitCustomAlpha(alpha);
```

`SetLpfMakeupGain8Bit()`

Set post-LPF gain for 8-bit audio.

```
void SetLpfMakeupGain8Bit(float gain);
```

Parameters:

- `gain` : Gain multiplier (e.g., 1.0 = no change, 1.08 = +8%)

Example:

```
SetLpfMakeupGain8Bit(1.15f); // Boost 8-bit audio by 15%
```

Status Accessors

`GetPlaybackState()`

Query current playback state (for non-blocking polling).

```
uint8_t GetPlaybackState(void);
```

Returns:

- `PB_Idle` , `PB_Playing` , `PB_Paused` , etc.

Example:

```
if (GetPlaybackState() == PB_Playing) {
 printf("Audio is playing...\n");
}
```

`GetPlaybackSpeed()`

Get current sample rate.

```
uint32_t GetPlaybackSpeed(void);
```

Hardware Integration Functions

These must be defined by the application to integrate the audio engine with your specific hardware.

`AudioEngine_DACSwitch()`

Function pointer to control amplifier GPIO (on/off).

```
extern DAC_SwitchFunc AudioEngine_DACSwitch;

// Application must define:
void MyDACControl(GPIO_PinState setting) {
 if (setting == SET) {
 gpio_bit_write(AMP_EN_GPIO_Port, AMP_EN_Pin, SET); // ON
 } else {
 gpio_bit_write(AMP_EN_GPIO_Port, AMP_EN_Pin, RESET); // OFF
 }
}

// In initialization:
AudioEngine_DACSwitch = MyDACControl;
```

`AudioEngine_ReadVolume()`

Function pointer to read volume setting (0–65535). Raw values are scaled as gain during playback. 0 is treated as 1 for minimum volume. Values are subject to the configured volume response curve (linear by default, or non-linear gamma curve if enabled).

```
extern ReadVolumeFunc AudioEngine_ReadVolume;

// Application must define:
uint16_t MyReadVolume(void) {
 // Read GPIO pins or ADC to determine volume level (0-65535)
 uint16_t volume = ReadMyVolumeSource();
 return volume ? volume : 1; // Treat 0 as 1 for minimum volume
}

// In initialization:
AudioEngine_ReadVolume = MyReadVolume;

// Optionally configure volume response curve:
SetVolumeResponseNonlinear(1); // Enable non-linear response (default)
SetVolumeResponseGamma(2.0f); // Set gamma exponent (1.0-4.0)
```

`AudioEngine_I2SInit()`

Function pointer to re-initialize I2S if needed (called after pause/resume).

```
extern I2S_InitFunc AudioEngine_I2SInit;

// Application must define:
void MyI2Init(void) {
```

```
 My_I2S_Init_code();
}

// In initialization:
AudioEngine_I2SInit = MyI2SInit;
```

DMA Callbacks

Connect these to your I2S DMA interrupt handlers:

```
// In your I2S interrupt service routine:
void I2S_TxHalfCpltCallback( void ) {
 // Called when first half of DMA buffer is transmitted
 // Audio engine processes next chunk
}

void I2S_TxCpltCallback( void ) {
 // Called when second half of DMA buffer is transmitted
}
```

The audio engine provides these implementations that will be called automatically.

Filter Configuration

16-bit LPF Aggressiveness Levels

The 16-bit biquad uses **lower α for heavier filtering** (same direction as the 8-bit one-pole). The coefficient formula `b0 = ((65536 - alpha) * (65536 - alpha)) >> 17` means lower alpha values result in more aggressive low-pass filtering.

Level	Alpha	Notes
Very Soft	0.625	Minimal filtering / brightest tone / highest cutoff
Soft	~0.80	Gentle filtering
Medium	0.875	Balanced filtering
Firm	~0.92	Firm filtering / lower cutoff
Aggressive	~0.97	Strongest filtering / darkest tone / lowest cutoff

- Warm-up (16 passes) still runs to suppress startup artefacts at the most aggressive setting.

Recommended Input Range for Best Quality:

The 16-bit biquad has feedback that can cause overshoot and ringing, especially at aggressive filter levels. To avoid clipping while preserving dynamic range:

Level	Recommended Range	Notes

LPF_VerySoft	75–85% of full scale (±24,500 to ±27,750)	Minimal overshoot risk
LPF_Soft	70–80% of full scale (±22,937 to ±26,214)	Good balance (recommended)
LPF_Medium	70–75% of full scale (±22,937 to ±24,500)	Increasing feedback
LPF_Firm	65–75% of full scale (+/-21,300 to +/-24,500)	Stronger feedback; moderate headroom
LPF_Aggressive	60–70% of full scale (±19,660 to ±22,937)	Strong feedback; conservative headroom essential

General Guideline:

Use **70–80% of full scale (±23,000)** as a safe starting point. If using LPF_Aggressive, stay closer to 70%; if using LPF_VerySoft, you can push toward 80–85%.

8-bit LPF Aggressiveness Levels

8-bit audio uses a **first-order (one-pole) filter** rather than a biquad. This architecture avoids feedback loop instability on quantized 8-bit data. As a result, the alpha range is narrower than the 16-bit biquad to maintain filter stability.

Filter Architecture:

- **One-pole formula**: `output = (α × input + (1 − α) × prev_output) × makeup_gain`
- **Why narrower range**: One-pole filters at low alpha (high filtering) can amplify quantization noise; biquads are more robust to this.

Level	Alpha	Cutoff Freq
Very Soft	0.9375	~3200 Hz
Soft	0.875	~2800 Hz
Medium	0.75	~2300 Hz
Firm	0.6875	~2000 Hz
Aggressive	0.625	~1800 Hz

Note on Range Differences:

The 16-bit biquad (α: 0.625 $\rightarrow$ 0.97) and 8-bit one-pole (α: 0.625 $\rightarrow$ 0.9375) do *not* span the same range. This is intentional: the biquad's wider range is safe for 16-bit data, while the one-pole's narrower range prevents instability on 8-bit input. Both filters provide LPF_VerySoft, LPF_Soft, LPF_Medium, LPF_Firm, and LPF_Aggressive presets for user consistency, but their underlying coefficients differ.

Important: The two filter types have the **same relationship** between alpha and filtering: **lower alpha = more filtering** for both architectures. The biquad coefficient formula uses `(65536 - alpha)` , which inverts the typical relationship.

DC Blocking Filter

Removes DC offset and very low frequencies.

Variant	Alpha	Cutoff Freq	Use
Standard	0.98	~44 Hz	Normal playback

Soft	0.995	~22 Hz	Gentler high-pass (use if ultra-low audio needed)

When to Enable `enable_soft_dc_filter_16bit` **:**

- Music with extended bass (< 44 Hz content)
- Subwoofer testing
- Normally leave disabled for typical speech/alerts

Soft Clipping Threshold

Soft clipping prevents harsh digital distortion when audio peaks exceed a threshold.

Configuration:

- **Threshold**: ±28,000 (85% of ±32,767 full scale)
- **Curve**: Cubic smoothstep ($s(x) = 3x^2 - 2x^3$)
- **Benefit**: Musical, transparent limiting

When to Enable:

- Always recommended (prevents clipping artefacts)
- Disable only if maximum undistorted headroom needed

Air Effect (High-Shelf Brightening Filter)

The Air Effect is an optional high-shelf filter that adds presence and brightness to audio by boosting high-frequency content. It uses a simple one-pole shelving architecture for CPU efficiency.

Configuration (defaults):

- **Type**: High-shelf one-pole filter
- **Shelf Gain (Q16)**: 98304 (~1.5×, ≈ +1.6 dB at Nyquist for α=0.75)
- **Shelf Gain Max (Q16)**: 131072 (2.0× cap to avoid harshness)
- **Cutoff Alpha**: 0.75 (~5–6 kHz shelving frequency @ 22 kHz)
- **Default State**: Disabled (`enable_air_effect = 0`)

Runtime Control (dB or Q16):

- `SetAirEffectGainDb(float db)` : set target HF boost in dB (computes Q16 internally, clamped to max)
- `GetAirEffectGainDb(void)` : read current boost in dB
- `SetAirEffectGainQ16(uint32_t gain_q16)` : set raw Q16 shelf gain (clamped)
- `GetAirEffectGainQ16(void)` : read raw Q16 shelf gain
- Presets (built-in): `{+1 dB, +2 dB, +3 dB}` with helpers:
- `SetAirEffectPresetDb(uint8_t preset_index)`
- `CycleAirEffectPresetDb(void)`
- `GetAirEffectPresetIndex/Count/GetAirEffectPresetDb`

Filter Characteristics:

The Air Effect works by separating high-frequency content and amplifying it:

1. Extract high-frequency component: `high_freq = input - prev_input`
2. Amplify high frequencies: `boost = high_freq × (1 − α) × shelf_gain`
3. Blend with smoothed output: `output = (α × input) + ((1 − α) × prev_output) + boost`

When to Enable:

- Muffled or dark-sounding samples → adds clarity and presence
- Quiet samples → adds energy and perceived loudness
- Archived audio → brightens aged or compressed recordings
- **Do not enable** if audio already sounds bright or harsh (risk of harshness)

Typical Use Case:

```
// Enable Air Effect and choose +2 dB preset
SetAirEffectPresetDb(2); // preset 0=off, 1=+1dB, 2=+2dB, 3=+3dB
// (Auto-disables if preset=0, auto-enables if preset>0)
SetLpf16BitLevel(LPF_Soft);

PlaySample(
 muffled_doorbell,
 sample_size,
 22000,
 16,
 Mode_mono
);

// Adjust live (e.g., button/UART):
CycleAirEffectPresetDb();
```

Filter Chain Order:

The Air Effect is positioned after the DC blocking filter but before fade/clipping effects:

```
16/8-bit LPF (optional: enable_16bit_biquad_lpf / enable_8bit_lpf)
 ↓
DC Blocking Filter (always on: standard or soft mode)
 ↓
AIR EFFECT (optional: enable_air_effect)
 ↓
Fade In/Out (always active)
 ↓
Noise Gate (optional: enable_noise_gate)
 ↓
Soft Clipping (optional: enable_soft_clipping, recommended)
 ↓
```

```
Volume Scaling (always active)
```

Tuning the Effect:

• For subtle brightening: use `SetAirEffectGainDb(1.0f)` or `SetAirEffectPresetDb(1)` (+1 dB).

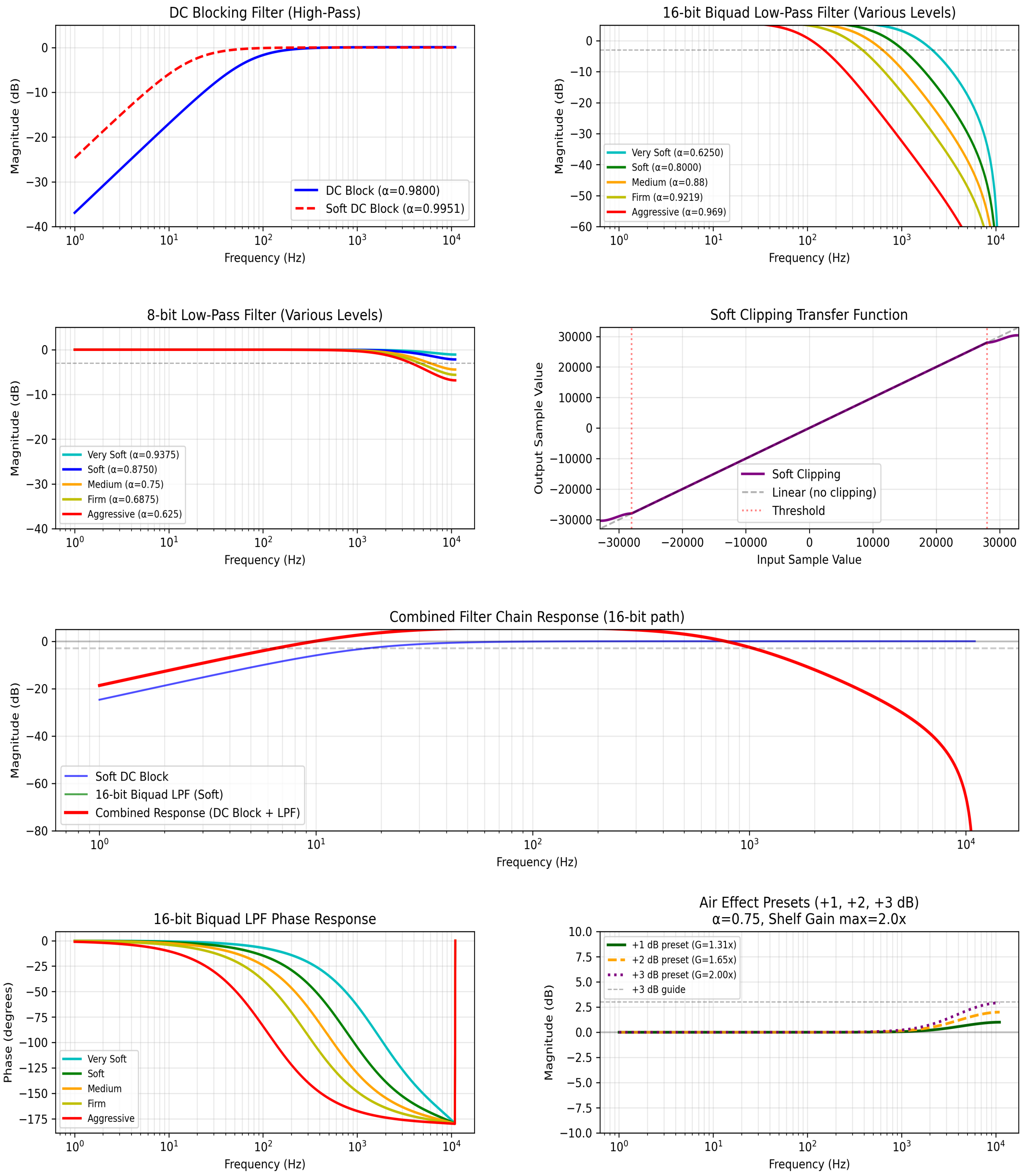
• For more sparkle: use `SetAirEffectGainDb(2.0f)` or `SetAirEffectPresetDb(2)` (+2 dB).

• For stronger presence: `SetAirEffectGainDb(3.0f)` or preset 3 (+3 dB).

• For a subtle lift: `SetAirEffectGainDb(0.0f)` (flat) or reduce gain below 0 dB if adding presence elsewhere.

• For different sample rates (e.g., 48 kHz), raise `AIR_EFFECT_CUTOFF` (higher α) to keep the shelf in the upper band.

Figure 1: Comprehensive Filter Frequency Response Analysis
Audio Engine DSP Filter Characteristics - Page 1: Filter Responses
GD32 Audio Engine @ 22 kHz Sample Rate



Playing Audio

Basic Playback Workflow

```c
#include "audio_engine.h"

// 1. Startup (once during initialization, e.g., in main.c)
static void AudioEngine_Init(void) {
 // Wire hardware hooks
 AudioEngine_DACSwitch = DAC_MasterSwitch;
 AudioEngine_ReadVolume = ReadVolume;
 AudioEngine_I2SInit = spi_config;

 // Configure filters
 FilterConfig_TypeDef cfg = filter_cfg; // start from defaults
 cfg.enable_16bit_biquad_lpf = 0;
 cfg.enable_8bit_lpf = 1;
 cfg.enable_soft_dc_filter_16bit = 0;
 cfg.enable_soft_clipping = 1;
 SetFilterConfig(&cfg);

 // Optional tuning
 SetLpf16BitLevel(LPF_Soft);
 SetAirEffectPresetDb(2); // +2 dB preset (auto-enables air effect)
}

// 2. Play an audio sample
static void PlayAlert(void) {
 extern const uint8_t alert_16bit_mono[];
 extern const uint32_t alert_16bit_mono_size;

 PB_StatusTypeDef result = PlaySample(
 alert_16bit_mono,
 alert_16bit_mono_size,
 22000, // Sample rate
 16, // 16-bit depth
 Mode_mono // Mono playback
 );

 if (result == PB_Playing) {
 WaitForSampleEnd();
 }
}
```

```
// 3. Non-blocking playback
static void PlayAlertNonBlocking(void) {
 PlaySample(alert_16bit_mono, alert_16bit_mono_size, 22000, 16, Mode_mono);
 // Returns immediately; playback happens in background
}

static void CheckPlaybackStatus(void) {
 if (GetPlaybackState() == PB_Playing) {
 printf("Still playing...\n");
 } else {
... (truncated)
```

Multi-Sample Playback Sequence

```
void PlayDoorbell(void) {
 // First: chime sound (16-bit, gentle filtering)
 SetLpf16BitLevel(LPF_Soft);
 PlaySample(chime_16bit, chime_size, 22000, 16, Mode_mono);
 WaitForSampleEnd();

 // Small delay between sounds
 delay_ms(500);

 // Second: bell sound (16-bit, medium filtering)
 SetLpf16BitLevel(LPF_Medium);
 PlaySample(bell_16bit, bell_size, 22000, 16, Mode_mono);
 WaitForSampleEnd();

 printf("Doorbell sequence complete\n");
}
```

Adjusting Playback on the Fly

```
void InteractivePlayback(void) {
 // Start playback with default settings
 SetLpf16BitLevel(LPF_Medium);
 PlaySample(my_audio, my_audio_size, 22000, 16, Mode_mono);

 while (GetPlaybackState() == PB_Playing) {
 // Monitor user input
 if (user_pressed_filter_button) {
 // Change filter level mid-playback
 SetLpf16BitLevel(LPF_Aggressive);
 }

 if (user_pressed_pause_button) {
 PausePlayback();
 }

 if (user_pressed_resume_button) {
 ResumePlayback();
 }

 delay_ms(100);
 }
}
```

Filter Parameters & Tuning

Understanding Alpha Coefficients

All filters in the audio engine use first-order or second-order IIR (infinite impulse response) filters with feedback coefficient α **(alpha)**.

First-Order Filter:

```
y[n] = α · x[n-1] + (1 - α) · y[n-1]
```

- α close to 1.0: Less filtering (high frequencies pass through)
- α close to 0.0: More filtering (stronger attenuation)

Biquad (Second-Order) Filter:

```
y[n] = b0·x[n] + b1·x[n-1] + b2·x[n-2] - a1·y[n-1] - a2·y[n-2]
```

Where coefficients are derived from α:

- $b0 = ((1 - \alpha)^2) / 2$

- $b1 = 2 \cdot b0$
- $b2 = b0$
- $a1 = -2 \cdot \alpha$
- $a2 = \alpha^2$

Warm-Up Behavior

Problem: With aggressive filtering (α = 0.625), the first playback sample causes a brief "cracking" sound due to the filter initializing from zero state.

Solution: Configurable Warm-Up (Default: 16 passes)

- Automatically invoked when playing 16-bit audio with enabled LPF
- Feeds the first audio sample through the biquad filter `BIQUAD_WARMUP_CYCLES` times on each channel (default: 16)
- Allows filter state to converge smoothly before DMA streaming starts
- Result: Eliminates startup transient artefacts

Configuration:

The warm-up behavior can be adjusted by changing the `BIQUAD_WARMUP_CYCLES` define in `audio_engine.h` :

```
#define BIQUAD_WARMUP_CYCLES 16 // Default: 16 passes (was 8)
```

Code Example (from audio_engine.c):

```
if (sample_depth == 16 && filter_cfg.enable_16bit_biquad_lpf) {
 int16_t first_sample = *((int16_t *)sample_to_play);
 // Run BIQUAD_WARMUP_CYCLES passes to let filter state settle
 for (uint8_t i = 0; i < BIQUAD_WARMUP_CYCLES; i++) {
 ApplyLowPassFilter16Bit(first_sample,
 &lpf_16bit_x1_left, &lpf_16bit_x2_left,
 &lpf_16bit_y1_left, &lpf_16bit_y2_left);
 ApplyLowPassFilter16Bit(first_sample,
 &lpf_16bit_x1_right, &lpf_16bit_x2_right,
 &lpf_16bit_y1_right, &lpf_16bit_y2_right);
 }
}
```

Q16 Fixed-Point Arithmetic

All filter coefficients and gains use **Q16 fixed-point representation**:

```
Q16 Value = Integer Value × 65536
Example: 1.0 = 65536 (0x10000)
 0.5 = 32768 (0x8000)
 1.08 ≈ 70779
```

Advantages:

- No floating-point hardware required (faster on MCU)
- Deterministic, no rounding surprises
- Easy to implement in assembly if needed

Converting Gain to Q16:

```
float gain = 1.08;
uint32_t gain_q16 = (uint32_t)(gain * 65536.0f); // 70779
SetLpfMakeupGain8Bit(gain); // Convenience function
```

Tuning Guide

For Speech/Alert Sounds

```
FilterConfig_TypeDef cfg = {
 .enable_16bit_biquad_lpf = 1,
 .enable_soft_dc_filter_16bit = 0, // Not needed for speech
 .enable_8bit_lpf = 1,
 .enable_noise_gate = 0, // Or 1 if background noise
 .enable_soft_clipping = 1,
 .lpf_makeup_gain_q16 = 70779, // 1.08x
 .lpf_16bit_level = LPF_Soft // Gentle, preserve clarity
};
SetFilterConfig(&cfg);
```

For Bass-Heavy Music

```
FilterConfig_TypeDef cfg = {
 .enable_16bit_biquad_lpf = 1,
 .enable_soft_dc_filter_16bit = 1, // Preserve low bass
 .enable_8bit_lpf = 1,
 .enable_noise_gate = 0,
 .enable_soft_clipping = 1,
 .lpf_makeup_gain_q16 = 65536, // 1.0x (no boost)
 .lpf_16bit_level = LPF_VerySoft // Minimal filtering
};
SetFilterConfig(&cfg);
```

For Noisy Environments

```
FilterConfig_TypeDef cfg = {
 .enable_16bit_biquad_lpf = 1,
 .enable_soft_dc_filter_16bit = 0,
 .enable_8bit_lpf = 1,
 .enable_noise_gate = 1, // Suppress low-level noise
 .enable_soft_clipping = 1,
 .lpf_makeup_gain_q16 = 70779,
 .lpf_16bit_level = LPF_Medium // Balanced noise reduction
};
SetFilterConfig(&cfg);
```

Volume Control

Overview

The audio engine applies volume scaling during sample processing. **Your application provides the volume input** by implementing a custom `ReadVolume()` function that returns a uint16_t value in the range 1–65535.

You can implement this function using various input methods:

- **Digital GPIO inputs** (OPT1–OPT3 pins for 3-bit binary encoding) — example pattern provided below
- **Analog ADC input** (12-bit potentiometer or variable resistance) — example pattern provided below
- **Other methods** (CAN bus, UART commands, network, etc.) — follow the same scaling approach

All implementations support **non-linear (logarithmic) volume response** to match human hearing perception.

Non-Linear Volume Response

Human hearing perceives loudness logarithmically, not linearly. The audio engine provides a configurable gamma-curve response.

Enable in main.h:

```
#define VOLUME_RESPONSE_NONLINEAR // Enable non-linear curve
#define VOLUME_RESPONSE_GAMMA 2.0f // Gamma exponent (typical value)
```

How it works:

- Linear input (1–65535) is normalized to 0.0–1.0
- Gamma curve is applied: `output = input^(1/gamma)`
- Result scales back to 1–65535 for audio attenuation

Gamma Values:

Gamma	Perception	Use Case
1.0	Linear	Reference (no curve)
2.0	Quadratic (recommended)	Most intuitive for human control
2.5	Stronger curve	Aggressive low-volume response

With **gamma = 2.0** (quadratic):

- **Low volumes** (0–50%): Small slider movement → big loudness change
- **High volumes** (50–100%): Big slider movement → small loudness change
- Result: More intuitive volume "feel" matching human perception

To disable and use linear scaling:

```
//#define VOLUME_RESPONSE_NONLINEAR // Comment this out
```

Volume Control Implementation

> **Note:** Volume input implementation is application-specific. The audio engine simply invokes the `ReadVolume()` callback and expects a uint16_t value in the range 1–65535. The application is responsible for:

> - Choosing the volume input method (digital GPIO, ADC potentiometer, CAN bus, UART command, etc.)

> - Scaling that input to the 1–65535 range

> - Handling debouncing, hysteresis, or other input conditioning

> - Being aware that if `VOLUME_RESPONSE_NONLINEAR` is enabled, the engine applies a gamma curve to the returned value

Application Example: Digital GPIO Volume Control

3-bit binary selector example (shows the pattern for application developers):

Selection	Binary	Scaled to 1–65535
Maximum	0b000	65535
75%	0b001	49151
50%	0b010	32767
...	...	...
Minimum	0b111	1

Example implementation in application (main.c):

```
uint16_t ReadVolume(void) {
 // Application reads three GPIO pins for volume
 uint8_t v =
 ( ( (OPT3_GPIO_Port->IDR & OPT3_Pin) != 0 ) << 2 ) |
 ( ( (OPT2_GPIO_Port->IDR & OPT2_Pin) != 0 ) << 1 ) |
 ( ( (OPT1_GPIO_Port->IDR & OPT1_Pin) != 0 ) << 0 );
```

```
 v = 7 - v; // Invert so 0b000 = max volume
 uint32_t scaled = ( (uint32_t)v * 65535U ) / 7U; // Map 0–7 to 1–65535
 return (uint16_t)scaled;
}
```

Application Example: Analog ADC Volume Control

Potentiometer example (shows the pattern for application developers):

Example implementation in application (main.c):

```
uint16_t ReadVolume(void) {
 // Application reads 12-bit ADC (0–4095)
 // Scale to 1–65535 range
 uint32_t lin = ((uint32_t)adc_raw * 65535U) / 4095U;
 return (uint16_t)lin;
}

void ADC0_1_IRQHandler( void )
{
 if( SET == adc_interrupt_flag_get(ADC1, ADC_INT_FLAG_EOC) ) {
 adc_interrupt_flag_clear(ADC1, ADC_INT_FLAG_EOC);
 adc_raw = adc_regular_data_read(ADC1);
 }
}
```

Hardware Integration

Application Integration Template

```c
#include "audio_engine.h"
#include <gd32f30x.h>

/* Hardware control functions (application-specific) */
void DAC_MasterSwitch(GPIO_PinState setting) {
 gpio_bit_write(AMP_EN_GPIO_Port, AMP_EN_Pin, setting);
}

uint16_t ReadVolume(void) {
 // Application-specific: read volume from GPIO, ADC, UART, etc.
 // Must return 1–65535 (0 is treated as 1)
 // The audio engine applies non-linear response if enabled

 // Example: 3-bit GPIO selector
 uint8_t v =
 ( ( (OPT3_GPIO_Port->IDR & OPT3_Pin) != 0 ) << 2 ) |
 ( ( (OPT2_GPIO_Port->IDR & OPT2_Pin) != 0 ) << 1 ) |
 ( ( (OPT1_GPIO_Port->IDR & OPT1_Pin) != 0 ) << 0 );
 v = 7 - v; // Invert: 0b000 = max volume
 uint32_t scaled = ( (uint32_t)v * 65535U ) / 7U;
 return (uint16_t)scaled;
}

/* Main initialization (in main.c startup sequence) */
void SystemInit_Audio(void) {
 // Set hardware callbacks before playing audio
 AudioEngine_DACSwitch = DAC_MasterSwitch;
 AudioEngine_ReadVolume = ReadVolume;
 AudioEngine_I2SInit = spi_config;

 // Configure filter settings (optional, defaults work for most cases)
 FilterConfig_TypeDef cfg;
 GetFilterConfig(&cfg);
 cfg.enable_soft_clipping = 1;
 SetFilterConfig(&cfg);

 printf("Audio engine ready\n");
}
```

Examples

Example 1: Simple Doorbell

```
#include "audio_engine.h"

// Audio data (define in flash)
extern const uint8_t doorbell_mono_16bit[];
extern const uint32_t doorbell_mono_16bit_size;

void PlayDoorbell(void) {
 SetLpf16BitLevel(LPF_Soft);
 PB_StatusTypeDef result = PlaySample(
 doorbell_mono_16bit,
 doorbell_mono_16bit_size,
 22000, // 22 kHz
 16, // 16-bit
 Mode_mono // Mono
 );

 if (result == PB_Playing) {
 WaitForSampleEnd();
 printf("Doorbell complete\n");
 } else {
 printf("Failed to play doorbell\n");
 }
}
```

Example 2: Multi-Tone Alert with Different Filters

```
void PlayAlert(void) {
 extern const uint8_t tone1[], tone2[], tone3[];
 extern const uint32_t tone1_size, tone2_size, tone3_size;

 // First tone: gentle
 SetLpf16BitLevel(LPF_VerySoft);
 PlaySample(tone1, tone1_size, 22000, 16, Mode_mono);
 WaitForSampleEnd();
 delay_ms(200);

 // Second tone: medium
 SetLpf16BitLevel(LPF_Medium);
 PlaySample(tone2, tone2_size, 22000, 16, Mode_mono);
 WaitForSampleEnd();
 delay_ms(200);

 // Third tone: aggressive (emphasis)
 SetLpf16BitLevel(LPF_Aggressive);
 PlaySample(tone3, tone3_size, 22000, 16, Mode_mono);
 WaitForSampleEnd();
}
```

Example 3: Voice Message with Configurable Filtering

```
void PlayVoiceMessage(LPF_Level filter_level) {
 extern const uint8_t message_16bit[];
 extern const uint32_t message_16bit_size;

 // Set filter before playback
 SetLpf16BitLevel(filter_level);

 PB_StatusTypeDef result = PlaySample(
 message_16bit,
 message_16bit_size,
 22000,
 16,
 Mode_mono
 );

 if (result == PB_Playing) {
 printf("Message playing with filter level %d\n", filter_level);
 }
}

// Usage:
// PlayVoiceMessage(LPF_Soft); // Clear
// PlayVoiceMessage(LPF_Aggressive); // Compressed
```

Example 4: Pause/Resume Functionality

```
volatile uint8_t pause_requested = 0;

void PlaybackTask(void) {
 SetLpf16BitLevel(LPF_Medium);
 PlaySample(my_audio, my_audio_size, 22000, 16, Mode_mono);

 while (GetPlaybackState() == PB_Playing) {
 if (pause_requested) {
 PausePlayback();
 printf("Paused\n");

 while (!resume_requested && GetPlaybackState() == PB_Paused) {
 delay_ms(50);
 }

 ResumePlayback();
 printf("Resumed\n");
 pause_requested = 0;
 resume_requested = 0;
 }

 delay_ms(50);
 }
}

// Button handler:
void EXTI_PauseButton_Handler(void) {
 pause_requested = 1;
}

void EXTI_ResumeButton_Handler(void) {
 resume_requested = 1;
}
```

Example 5: Non-Blocking Playback with Status Checking

```
void NonBlockingPlayback(void) {
 // Start playing
 SetLpf16BitLevel(LPF_VerySoft);
 PlaySample(background_music, bg_music_size, 22000, 16, Mode_stereo);

 // Do other work while audio plays
 for (int i = 0; i < 100; i++) {
 if (GetPlaybackState() == PB_Playing) {
 printf("Playing: %d%% complete\n", (i+1));
 } else {
 printf("Playback finished\n");
 break;
 }

 delay_ms(100); // 10 second total wait
 }
}
```

Example 6: Accessibility — Filter Settings for Hard of Hearing

This example demonstrates filter configuration optimized for users with hearing loss, emphasizing speech clarity and presence without over-filtering.

```
void SetAccessibleAudio(void) {
 // Configuration optimized for hearing-impaired listeners
 // Focus: speech clarity and presence in 2-6 kHz band
 FilterConfig_TypeDef cfg = {
 .enable_16bit_biquad_lpf = 1,
 .lpf_16bit_level = LPF_Soft, // Gentle filtering preserves clarity
 .enable_soft_dc_filter_16bit = 1, // Softer DC removal (22 Hz cutoff)
 .enable_8bit_lpf = 1,
 .enable_noise_gate = 0, // Keep quiet consonants (s, th, sh)
 .enable_soft_clipping = 1, // Reduce harsh peaks
 .enable_air_effect = 1, // Boost presence in 2-6 kHz
 .lpf_makeup_gain_q16 = 82000 // ~1.25x gain (Q16 fixed-point)
 };

 SetFilterConfig(&cfg);
 SetAirEffectPresetDb(2); // +2 dB presence boost (mid-range)
}

// Usage in doorbell application
void play_accessible_alert(void) {
```

```
 SetAccessibleAudio();


 PlaySample(alert_tone, alert_size, 22000, 16, Mode_mono);
 WaitForSampleEnd();
}
```

Design Rationale:

- **LPF_Soft** (α ≈ 0.80) — Gentler than Medium/Aggressive; prevents over-smoothing that muddies speech
- **No Noise Gate** — Preserves subtle consonants and quiet speech details critical for comprehension
- **Air Effect +2 dB** — Compensates for typical age-related high-frequency loss; boosts presence band (2–6 kHz where speech consonants live)
- **Makeup Gain ~1.25x** — Offsets the LPF attenuation, maintaining perceived loudness
- **Soft DC Filter** — Gentler transition than standard DC blocking; avoids unnatural clicks

Alternative for Severe Loss:

```
// For users with more pronounced loss, use Very Soft + stronger presence
cfg.lpf_16bit_level = LPF_VerySoft; // Lightest filtering
SetFilterConfig(&cfg);
SetAirEffectPresetDb(3); // +3 dB (strongest preset)
```

Troubleshooting

Issue: Audio Not Playing

Symptoms: `PlaySample()` returns `PB_Error` or `PB_PlayingFailed`

Solutions:

1. Verify I2S is initialized and `AudioEngine_I2SInit` points to `spi_config`
2. Check that `AudioEngine_I2SInit` callback is set
3. Confirm DMA is enabled for I2S2 TX
4. Check amplifier GPIO is working: `gpio_bit_write(AMP_EN_GPIO_Port, AMP_EN_Pin, SET)`
5. Verify audio data pointer is valid (in flash or accessible RAM)

Debug:

```
uint8_t state = GetPlaybackState();
printf("Playback state: %d\n", state); // 0=Idle, 2=Playing, etc.


SetLpf16BitLevel(LPF_Soft);
PB_StatusTypeDef result = PlaySample(my_audio, my_size, 22000, 16, Mode_mono);
printf("PlaySample result: %d\n", result);
```

Issue: Audio Playing but Distorted or Crackling

Symptoms: Output contains harsh noise or crackles, especially at start

Solutions:

1. **Enable soft clipping:**

```
FilterConfig_TypeDef cfg;
 GetFilterConfig(&cfg);
 cfg.enable_soft_clipping = 1;
 SetFilterConfig(&cfg);
```

2. **Adjust filter level to reduce aggressive processing:**

```
SetLpf16BitLevel(LPF_Medium); // Instead of LPF_Aggressive
```

3. **Reduce volume:**
   - Check that hardware volume setting (GPIO or analog input) is at reasonable level
   - Audio data may have peaks at or near ±32,767

4. **Verify audio data:**
   - Check that 16-bit samples are properly formatted (little-endian on ARM)
   - Ensure sample rate matches playback speed

Issue: Filter Sounds Too Thin/Bright

Symptoms: High frequencies dominate, lacks bass/warmth

Solutions:

```
// Increase filtering aggressiveness
SetLpf16BitLevel(LPF_Aggressive);

// Or enable soft DC blocking for extended low end
FilterConfig_TypeDef cfg;
GetFilterConfig(&cfg);
cfg.enable_soft_dc_filter_16bit = 1; // 22 Hz instead of 44 Hz
SetFilterConfig(&cfg);
```

Issue: Filter Sounds Too Muffled/Dull

Symptoms: High frequencies are suppressed, loss of clarity

Solutions:

```
// Reduce filtering aggressiveness
SetLpf16BitLevel(LPF_VerySoft);

// Or disable some filters entirely
FilterConfig_TypeDef cfg;
```

```
GetFilterConfig(&cfg);
cfg.enable_8bit_lpf = 0; // If playing 16-bit audio
SetFilterConfig(&cfg);
```

Issue: 8-bit Audio Sounds Noisy

Symptoms: Audible quantization noise from 8-bit samples

Solutions:

1. **Enable TPDF dithering (automatic)** - should already be on by default
2. **Increase makeup gain:**

```
SetLpfMakeupGain8Bit(1.15f); // Boost by 15%
```

3. **Use 16-bit audio if available** - much better quality

Issue: Memory Corruption / Hard Fault

Symptoms: Microcontroller resets or freezes during audio playback

Common Causes:

1. **Audio buffer pointer is invalid or not in accessible memory**
   - Audio data must be in flash (constant) or main RAM
   - Not in stack-only RAM

2. **DMA configuration issue**
   - Verify DMA word width is 32-bit (not 8 or 16)
   - Check DMA direction is I2S TX (transmit)

3. **I2S interrupt interfering with audio processing**
   - Ensure DMA IRQ handlers call `I2S_TxHalfCpltCallback()` / `I2S_TxCpltCallback()`
   - Verify DMA interrupt priorities don't conflict
   - **GD32 audio engine is SysTick-independent for stop/delay paths**. SysTick priority is no longer a lockup requirement for playback stop.
   - Keep DMA/I2S IRQ service latency low enough to avoid playback buffer underruns.

Debug:

```
// Add validation before playing
if ((uint32_t)audio_ptr < 0x08000000 && (uint32_t)audio_ptr >= 0x0A000000) {
 printf("Invalid audio pointer: 0x%08X\n", (uint32_t)audio_ptr);
}
```

Performance Notes

CPU Load

- **Processing overhead:** < 5% @ 22 kHz with full filter chain enabled
- **Per-sample time:** ~50 CPU cycles for all filters
- **DMA-driven:** Most processing offloaded from main CPU loop

Memory Usage

```
Flash (.text/.rodata, Release): ~12.9 KB (audio engine + filters + Air Effect presets
 )
RAM: ~2.5 KB (state variables + playback buffer)
```

Latency

- **Playback latency:** ~93 ms (2048 samples @ 22 kHz)
- 50 ms for DMA buffer
- 43 ms for warm-up and initial processing
- **Pause/Resume:** Immediate (within one DMA block, ~45 ms)

Quality Metrics

Metric	Value	Notes
Sample Rate	22 kHz	Nyquist: 11 kHz
Bit Depth	16-bit (native)	8-bit with TPDF dithering
Dynamic Range	96 dB (16-bit)	48 dB (8-bit effective)
SNR (w/ dithering)	102 dB (8-bit)	TPDF reduces quantization noise
THD (soft clipping)	< 0.1%	Cubic smoothstep minimizes distortion

Power Consumption

- **GD32F303CGT6 (typical):** ≤40 mA (core + peripherals)
- **I2S + DMA Active:** ~10 mA additional
- **Amplifier (MAX98357A):** ~100 mA @ 0.5W output, >1W capable
- **Total System @ 0.5W audio:** ~150 mA @ 5V
- **Total System @ 1W audio:** ~200 mA @ 5V

Note: The MAX98357A amplifier can deliver over 1W to an 8Ω speaker, significantly exceeding the GD32's typical current draw.

Summary

The Audio Engine provides a complete, production-ready solution for embedded audio playback with professional DSP filtering. Key design principles:

1. **Fixed-Point Integer Math** - No FPU required, deterministic on MCU
2. **Modular Filter Chain** - Enable/disable each stage independently
3. **Runtime Configuration** - Adjust filter parameters without recompilation
4. **DMA-Driven Streaming** - Efficient background audio playback
5. **Warm-Up Initialization** - Eliminates startup artefacts with aggressive filtering

For questions or issues, refer to the troubleshooting section or review the provided code examples.

Document Version: 1.0

Last Updated: 2026-03-01

Audio Engine Version: Modularized with Widened 16-bit LPF & Warm-Up Support